

A photograph of two women in a room. In the foreground, a woman with short blonde hair is smiling and looking towards the right. In the background, a woman with dark curly hair is also smiling and looking towards the left. The wall behind them is covered with several framed photographs of various scenes. The date '30 Apr 2025' is overlaid in the top right corner.

30 Apr 2025

Fraudulent Claim Detection Summary & Recommendations

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Problem Statement & Business Objectives

- Problem Statement
 - Global Insure faces heavy losses due to delayed fraud detection
 - Current manual process is slow and inefficient
 - Goal: Early classification of claims as fraudulent or legitimate
 - Objective: Minimize financial losses and optimize claim handling
- Business Objectives
 - Build a model using historical claims and customer profiles
 - Predict fraud likelihood before claims are approved
 - Identify key fraud predictors and derive actionable insight

Approach and Data Preparation

- Loaded and inspected data
- Removed missing and illogical values (e.g., negative losses)
- Dropped high-cardinality identifiers
- Categorical encoding and feature scaling applied
- Class imbalance resolved using RandomOverSampler

Exploratory Data Analysis – Key Insights

- Fraud more likely for certain occupations and hobbies (e.g., chess, cross-fit)
- Major damage incidents show 60%+ fraud likelihood
- Correlated features: claim amounts, vehicle & injury claims
- Males and JD/MBA holders more likely to report fraud

Feature Engineering

- Created binary indicators for risky segments
- Combined rare categories to reduce sparsity
- Applied one-hot encoding for categorical features
- Used RandomOverSampler to balance classes

Model Evaluation

Logistics Regression

- Feature selection via RFECV
- Optimal threshold of 0.4 selected via ROC
- Accuracy: ~86%, Precision: ~83%, Recall: ~89%, F1 Score: ~0.86
- Interpretable model with statistically significant features

Random Forest

- Feature importance used to select top predictors
- Hyperparameter tuning using GridSearchCV
- Accuracy: ~91%, Precision: ~87%, Recall: ~89%, F1 Score: ~0.89
- Outperformed logistic regression on all metrics

Recommendations & Business Implications

- Fraudulent claims exhibit identifiable patterns linked to incident severity, customer hobbies, and occupation types, making predictive modelling highly viable.
- The Random Forest model is recommended for operational deployment due to its superior accuracy (~91%), balanced precision and recall, and robustness after hyperparameter tuning.
- Although the Logistic Regression model showed slightly lower performance (~86% accuracy), it remains valuable for its interpretability, especially in regulatory and audit contexts.
- Prioritize investigation for high-risk segments (e.g., major damage, unusual hobbies)
- Enhance claims triage using predictive scores
- Reduce financial losses and processing delays significantly

A woman with dark curly hair and glasses is smiling and looking off to the side. The image is partially covered by a purple and pink gradient overlay at the bottom.

THANK YOU!